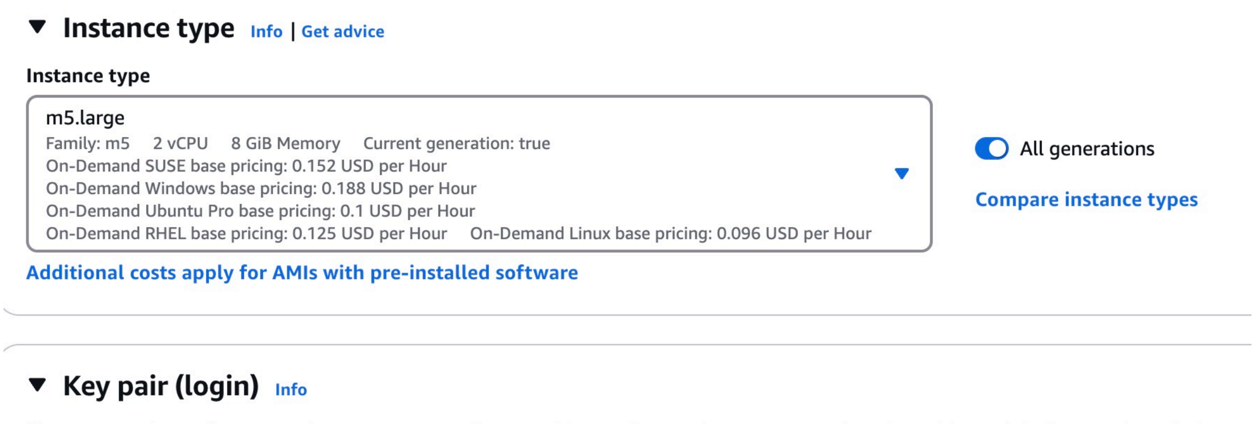
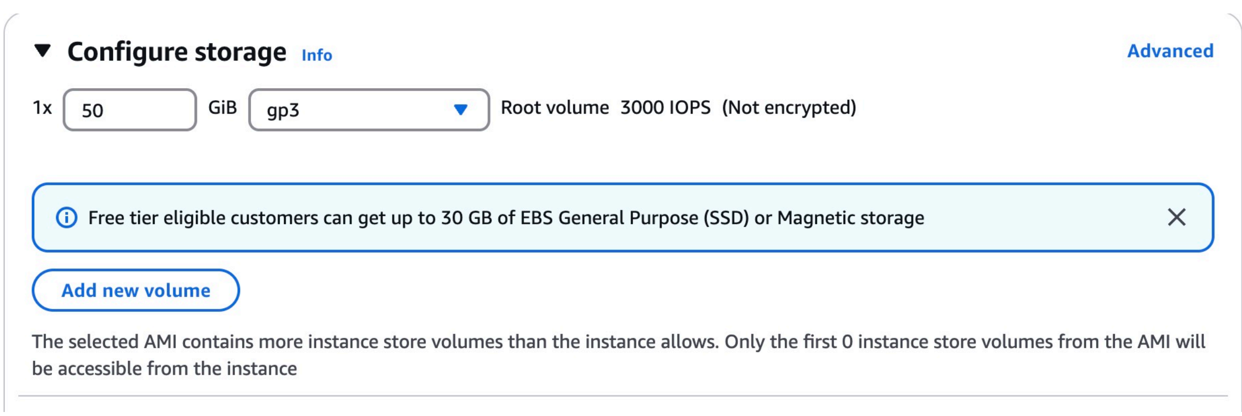
Assignment 5

Task1

Part1: Create Instance and see instance type and EBS

For Task1 to create EC2 instance, I was following the document and chose the m5.large as instance type, and chose the storage as 50GB





However, I found AWS academy is limited for the storage, even when I changed the storage under 30GB, it still doesn’t work, means I need to change the instance type as well.

Creating instance failed as follows:

0 x File systems Edit

▶ **Advanced details** Info

You are not authorized to perform this operation. User: arn:aws:sts::354754152263:assumed-role/voclabs/user3768455=Wang,_Ziyan is not authorized to perform: ec2:RunInstances on resource: arn:aws:ec2:us-east-1:354754152263:instance/* with an explicit deny in an identity-based policy. Encoded authorization failure message: VMJiN6BpfRcTzQRYZiJ9-qw131xnAC4mVoOmJn4Vd4rFah1Ar5dQz_BkfB95LwCN4s5HRO5lGkzn9lQ3-jOqS4DJHF2i8xaeXULB5GPpizUQpJgVYGIl_KKhZRAAQMBV8qAlZokRJTwssWd1tmfSryUhMTtZJZYlzm2pBCMwyGYfGi7LkNJitcEheBpsFz41neDg6gt4-7WMryifhfnnKlQr6qSv1M99phfFvnFIrbRfjnUZy37uCXRQOmVp4cPXlH82vJLNPt2iCRh6oZhkQKXJBO5ufeoIUUKWX1LDCELqJ-bVYiJoVHAw-1hK6KEhrtJHSFGWs8GFSS7nod3mHYiYo4_iYs6LgwyERroK0_MBdFzRZLIvFl4rdqxcCBG729D2tm0WU9kRP-D0lgoRBRB1DrnOBCGk0WUg8zS8GPA-S2bIOP1EuyZqwc9rckKQQeQFAut3cYTS-GMLf0cPXUludieocDg19_NXFKhmnRAZ8bnIrnHBp3gnKmoMYkOL3KyDwZ1bSuSCsGix9qkJxRKCyp1_JopLounHBVZR4TOlhHbsusOPIjGMUdxzIs8PgMCYR2QWUY90-u1HFTc1SiaDmHGphbQbOMYN5JdWOxsZexfScBsnu4SjCU5z0uk-chnPeM8m0HUxcQ36eM11h0lmK9CuUXNkQsjFB14YclVABFqiuUdC-bOTWctLu4NMutm6M7GgPIDH_6D2Y7Lqm-cYPrdSd2I-yHuW2GqpIYV96vWMoAe-JyDXGznL3fFqxRqjJtKiefbbPcSFyTtLsXRyI6VjHlFOZA_pk2rPBuDOHauZ-cyXtq12R-9wv4GN5a-LOADWsjkf2PAq9lroBb0UDzWZejyis-3ClcGGM39DLIQAYY3ZAkQIsOzXE6pwh30TJXPw0Yl3ze27i4MUHIFbZnJmW_UFPBst49wnM5FDrG6CHEVNlAak

Diagnose with Amazon Q

Instance launch failed

You are not authorized to perform this operation. User: arn:aws:sts::354754152263:assumed-role/voclabs/user3768455=Wang,_Ziyan is not authorized to perform: ec2:RunInstances on resource: arn:aws:ec2:us-east-1:354754152263:instance/* with an explicit deny in an identity-based policy. Encoded authorization failure message: VMJiN6BpfRcTzQRYZiJ9-qw131xnAC4mVoOmJn4Vd4rFah1Ar5dQz_BkfB95LwCN4s5HRO5lGkzn9lQ3-jOqS4DJHF2i8xaeXULB5GPpizUQpJgVYGIl_KKhZRAAQMBV8qAlZokRJTwssWd1tmfSryUhMTtZJZYlzm2pBCMwyGYfGi7LkNJitcEheBpsFz41neDg6gt4-7WMryifhfnnKlQr6qSv1M99phfFvnFIrbRfjnUZy37uCXRQOmVp4cPXlH82vJLNPt2iCRh6oZhkQKXJBO5ufeoIUUKWX1LDCELqJ-bVYiJoVHAw-1hK6KEhrtJHSFGWs8GFSS7nod3mHYiYo4_iYs6LgwyERroK0_MBdFzRZLIvFl4rdqxcCBG729D2tm0WU9kRP-D0lgoRBRB1DrnOBCGk0WUg8zS8GPA-S2bIOP1EuyZqwc9rckKQQeQFAut3cYTS-GMLf0cPXUludieocDg19_NXFKhmnRAZ8bnIrnHBp3gnKmoMYkOL3KyDwZ1bSuSCsGix9qkJxRKCyp1_JopLounHBVZR4TOlhHbsusOPIjGMUdxzIs8PgMCYR2QWUY90-u1HFTc1SiaDmHGphbQbOMYN5JdWOxsZexfScBsnu4SjCU5z0uk-chnPeM8m0HUxcQ36eM11h0lmK9CuUXNkQsjFB14YclVABFqiuUdC-bOTWctLu4NMutm6M7GgPIDH_6D2Y7Lqm-cYPrdSd2I-yHuW2GqpIYV96vWMoAe-JyDXGznL3fFqxRqjJtKiefbbPcSFyTtLsXRyI6VjHlFOZA_pk2rPBuDOHauZ-cyXtq12R-9wv4GN5a-LOADWsjkf2PAq9lroBb0UDzWZejyis-3ClcGGM39DLIQAYY3ZAkQIsOzXE6pwh30TJXPw0Yl3ze27i4MUHIFbZnJmW_UFPBst49wnM5FDrG6CHEVNlAak

Diagnose with Amazon Q

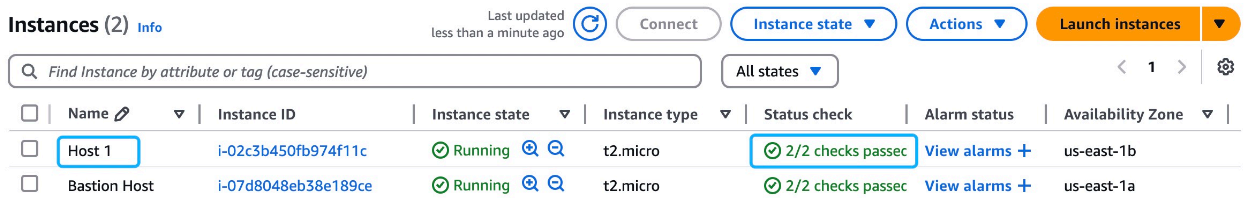
▼ **Launch log**

Initializing requests	Succeeded
Creating security groups	Succeeded
Creating security group rules	Succeeded
Launch initiation	Failed

Cancel Edit instance config Retry failed tasks

→ Solution: So I back to the Edit instance config and I changed the instance type to t2.miro, and storage was using the default 8GB.

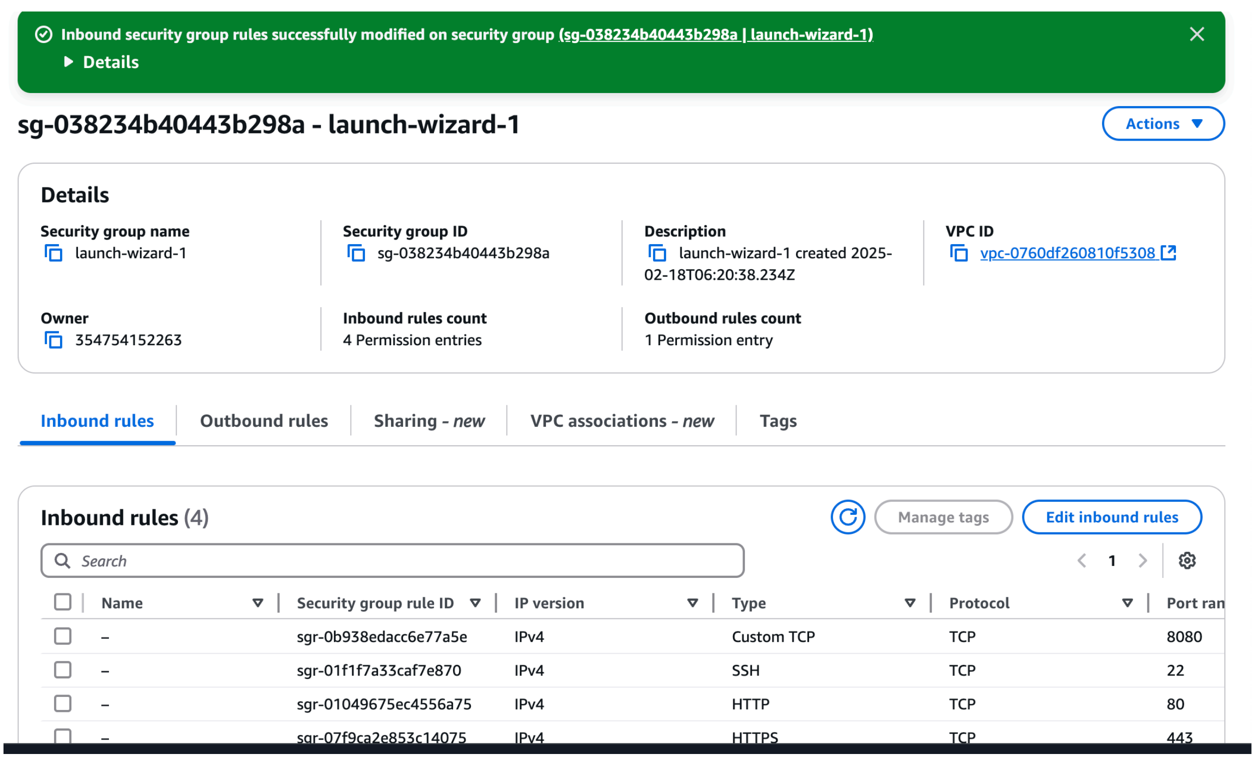
After changed, Instance created successfully:



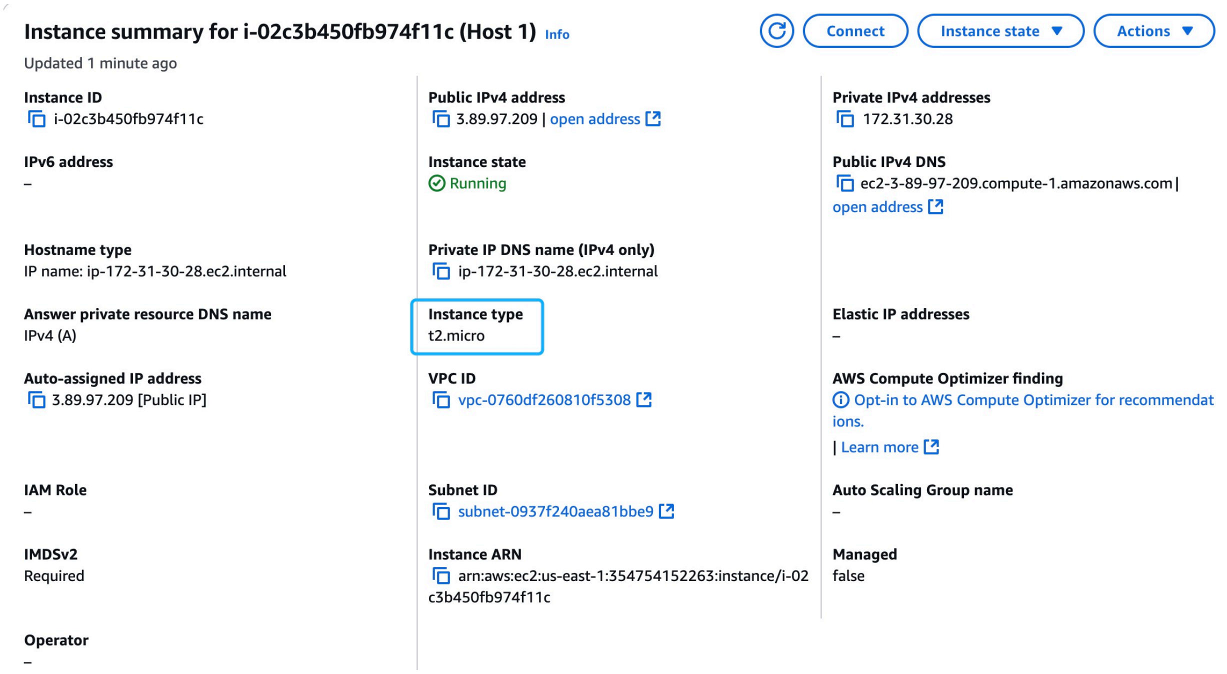
In security group, I add rules to the inbound rules:

- Port 22 for SSH.
- Port 80 and 443 for HTTP/HTTPS (if you plan to run web applications).
- Port 8080 for other services (like Portainer for Docker monitoring)

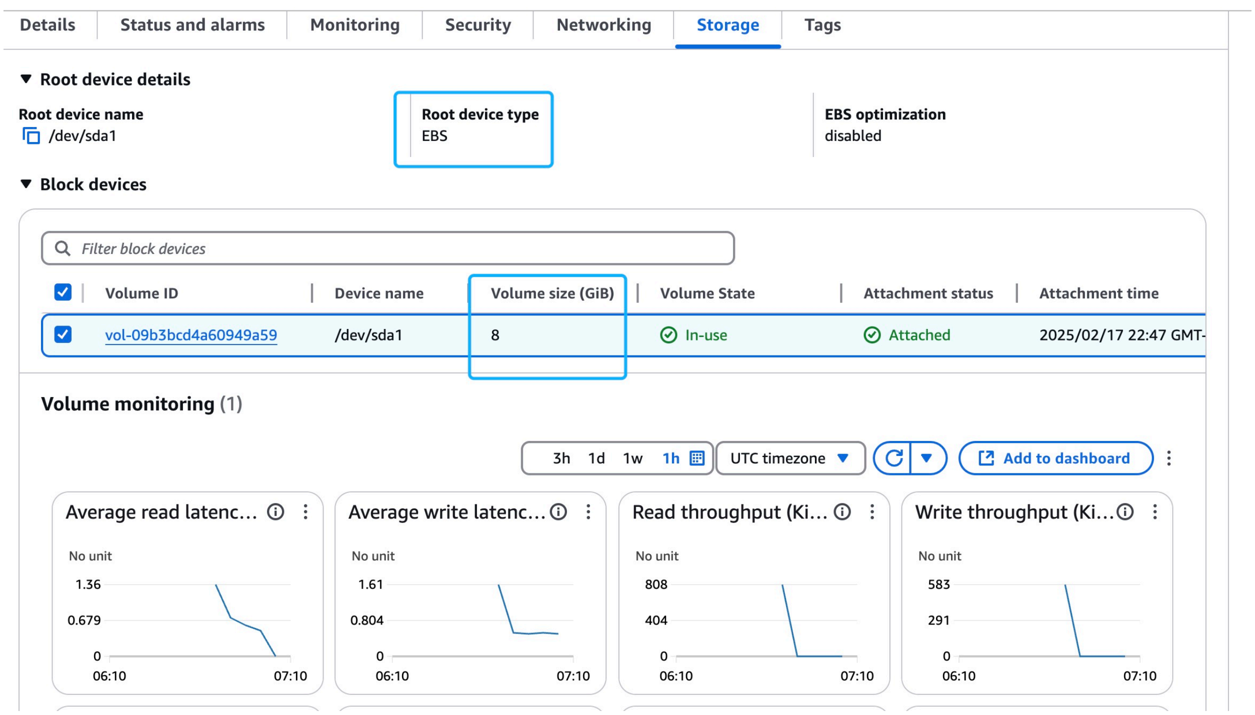
8080 is used for later open my web browser



Enter into the instance and see the instance type is t2.micro:



In storage, see the EBS is 8GB



Part2: install docker

```
Reading package lists... Done
W: GPG error: https://download.docker.com/linux/ubuntu noble InRelease: The foll
owing signatures couldn't be verified because the public key is not available: N
O_PUBKEY 7EA0A9C3F273FCD8
E: The repository 'https://download.docker.com/linux/ubuntu noble InRelease' is
not signed.
N: Updating from such a repository can't be done securely, and is therefore disa
bled by default.
N: See apt-secure(8) manpage for repository creation and user configuration deta
ils.
ubuntu@ip-172-31-30-28:~$ sudo apt-get install -y docker-ce
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Package docker-ce is not available, but is referred to by another package.
This may mean that the package is missing, has been obsoleted, or
is only available from another source

E: Package 'docker-ce' has no installation candidate
ubuntu@ip-172-31-30-28:~$ sudo docker --version
sudo: docker: command not found
ubuntu@ip-172-31-30-28:~$ dpkg -l | grep docker
ubuntu@ip-172-31-30-28:~$ which docker
ubuntu@ip-172-31-30-28:~$
```

I followed the document steps but failed to install docker.

By using command: 1. sudo docker --version 2.dpkg -l | grep docker 3.which docker, all of these commands didn’t have any output, means I didn’t install Docker successfully.

```
Reading package lists... Done
W: GPG error: https://download.docker.com/linux/ubuntu noble InRelease: The foll
owing signatures couldn't be verified because the public key is not available: N
O_PUBKEY 7EA0A9C3F273FCD8
E: The repository 'https://download.docker.com/linux/ubuntu noble InRelease' is
not signed.
N: Updating from such a repository can't be done securely, and is therefore disa
bled by default.
N: See apt-secure(8) manpage for repository creation and user configuration deta
ils.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Package docker-ce is not available, but is referred to by another package.
This may mean that the package is missing, has been obsoleted, or
is only available from another source

E: Package 'docker-ce' has no installation candidate
ubuntu@ip-172-31-30-28:~$
```

I was seeing this issue casue by: 1.NO_PUBKEY 7EA0A9C3F273FCD8 2.Ubuntu version (noble) currently does not support Docker official repository

So I removed the old repo first, then add Docker official GPG Key

```
ubuntu@ip-172-31-30-28:~$ sudo rm -rf /etc/apt/sources.list.d/docker.list
ubuntu@ip-172-31-30-28:~$ sudo apt-get update
sudo apt-get install -y ca-certificates curl
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Get:5 https://download.docker.com/linux/ubuntu noble InRelease [48.8 kB]
Err:5 https://download.docker.com/linux/ubuntu noble InRelease
  The following signatures couldn't be verified because the public key is not av
ailable: NO_PUBKEY 7EA0A9C3F273FCD8
Reading package lists... Done
W: GPG error: https://download.docker.com/linux/ubuntu noble InRelease: The foll
owing signatures couldn't be verified because the public key is not available: N
O_PUBKEY 7EA0A9C3F273FCD8
E: The repository 'https://download.docker.com/linux/ubuntu noble InRelease' is
not signed.
N: Updating from such a repository can't be done securely, and is therefore disa
bled by default.
N: See apt-secure(8) manpage for repository creation and user configuration deta
ils.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

Then use the Ubuntu 22.04 (Jammy) Docker repo

```
ubuntu@ip-172-31-30-28:~$ sudo install -m 0755 -d /etc/apt/keyrings
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo tee /etc/apt/keyr
ings/docker.asc > /dev/null
sudo chmod a+r /etc/apt/keyrings/docker.asc
ubuntu@ip-172-31-30-28:~$ echo "deb [arch=amd64 signed-by=/etc/apt/keyrings/dock
er.asc] https://download.docker.com/linux/ubuntu jammy stable" | sudo tee /etc/a
pt/sources.list.d/docker.list > /dev/null
ubuntu@ip-172-31-30-28:~$ sudo apt-get update
sudo apt-get install -y docker-ce docker-ce-cli containerd.io
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Get:5 https://download.docker.com/linux/ubuntu noble InRelease [48.8 kB]
Get:6 https://download.docker.com/linux/ubuntu jammy InRelease [48.8 kB]
Err:5 https://download.docker.com/linux/ubuntu noble InRelease
  The following signatures couldn't be verified because the public key is not av
ailable: NO_PUBKEY 7EA0A9C3F273FCD8
Get:7 https://download.docker.com/linux/ubuntu jammy/stable amd64 Packages [44.1
 kB]
Reading package lists... Done
```

Finally, update and install docker, and check if docker is installed successfully

```
ubuntu@ip-172-31-30-28:~$ sudo systemctl enable docker
sudo systemctl start docker
sudo docker --version
Synchronizing state of docker.service with SysV service script with /usr/lib/sys
temd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable docker
Docker version 27.5.1, build 9f9e405
ubuntu@ip-172-31-30-28:~$ sudo docker --version
Docker version 27.5.1, build 9f9e405
ubuntu@ip-172-31-30-28:~$
```

Yes, docker installed successfully, we can see the version now

Setp 2: Install Docker Compose

```
ubuntu@ip-172-31-30-28:~$ sudo curl -L "https://github.com/docker/compose/releas
es/latest/download/docker-compose-$(uname -s)-$(uname -m)" -o /usr/local/bin/doc
ker-compose
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0
100 70.1M  100 70.1M    0     0  65.0M      0  0:00:01  0:00:01 --:--:--  112M
ubuntu@ip-172-31-30-28:~$ sudo chmod +x /usr/local/bin/docker-compose
ubuntu@ip-172-31-30-28:~$ docker-compose --version
Docker Compose version v2.33.0
ubuntu@ip-172-31-30-28:~$
```

By seeing the version, means the docker-compose is installed successfully.

Task2: Setting Up a Simple Web Application with Nginx and MySQL Using Docker

using nano to open docker-compose.yml file and add configuration:

```
  GNU nano 7.2                         docker-compose.yml
version: '3.8'
services:
  web:
    image: nginx:latest
    ports:
      - "8080:80"
    volumes:
      - ./html:/usr/share/nginx/html # Bind mount the local html directory
  db:
   image: mysql:5.7
   environment:
    MYSQL_ROOT_PASSWORD: example
    MYSQL_DATABASE: mydb




                               [ Read 13 lines ]
^G Help       ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location
^X Exit       ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/ Go To Line
```

Editing HTML file by nano, and add my name Ziyan under h1 in HTML file:

```
 GNU nano 7.2                          html/index.html *
apiVersion: v1
kind: ConfigMap
metadata:
        name: web-content
data:
        index.html: |
                <!DOCTYPE html>
                <html lang="en">
                <head>
                        <meta charset="UTF-8">
                        <meta name="viewport" content="width=device-width,
initial-scale=1.0">
                        <title>Simple Web App</title>
                </head>
                <body>
                        <h1>Welcome to My Simple Web Application! - Ziyan</h1>
                        <p>This application is powered by Kubernetes and MySQL.>
                </body>
                </html>

^G Help       ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit       ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line
```

```
ziyanw — ubuntu@ip-172-31-30-28: ~/docker-web-mysql-app — ssh -i ~/...
 ✔ db Pulled                                                              15.4s
   ✔ 20e4dcae4c69 Pull complete                                            7.0s
   ✔ 1c56c3d4ce74 Pull complete                                            7.1s
   ✔ e9f03a1c24ce Pull complete                                            7.2s
   ✔ 68c3898c2015 Pull complete                                            7.7s
   ✔ 6b95a940e7b6 Pull complete                                            7.8s
   ✔ 90986bb8de6e Pull complete                                            7.8s
   ✔ ae71319cb779 Pull complete                                            9.7s
   ✔ ffc89e9dfd88 Pull complete                                            9.7s
   ✔ 43d05e938198 Pull complete                                           14.9s
   ✔ 064b2d298fba Pull complete                                           14.9s
   ✔ df9a4d85569b Pull complete                                           14.9s
[+] Running 3/3
 ✔ Network docker-web-mysql-app_default  Created                           0.1s
 ✔ Container docker-web-mysql-app-web-1  Started                           1.3s
 ✔ Container docker-web-mysql-app-db-1   Started                           1.3s
ubuntu@ip-172-31-30-28:~/docker-web-mysql-app$ sudo docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED         STATUS
      PORTS                                   NAMES
cd35c8c6711a   nginx:latest   "/docker-entrypoint.…"   6 seconds ago   Up 5 seco
nds   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   docker-web-mysql-app-web-1
20de9e4c87ff   mysql:5.7      "docker-entrypoint.s…"   6 seconds ago   Up 5 seco
nds   3306/tcp, 33060/tcp                     docker-web-mysql-app-db-1
ubuntu@ip-172-31-30-28:~/docker-web-mysql-app$
```

By running sudo docker ps, I can see both nginx and mysql are running.

Test the Application

I used to ways to test my app:

1. **Navigate to My web browser:**

http://3.89.97.209:8080

my name is also shown since I edited the html file

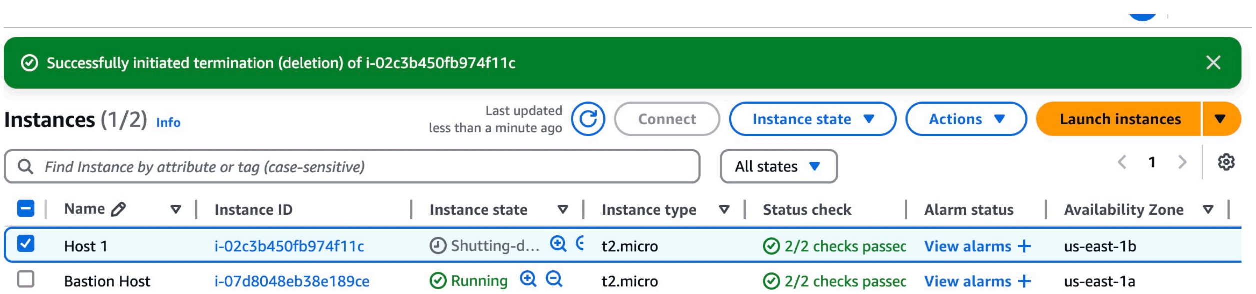
2. **use curl to test the endpoint**

```
ubuntu@ip-172-31-30-28:~/docker-web-mysql-app$ curl -v http://3.89.97.209:8080
*   Trying 3.89.97.209:8080...
* Connected to 3.89.97.209 (3.89.97.209) port 8080
> GET / HTTP/1.1
> Host: 3.89.97.209:8080
> User-Agent: curl/8.5.0
> Accept: */*
>
< HTTP/1.1 200 OK
< Server: nginx/1.27.4
< Date: Tue, 18 Feb 2025 08:02:23 GMT
< Content-Type: text/html
< Content-Length: 420
< Last-Modified: Tue, 18 Feb 2025 07:52:46 GMT
< Connection: keep-alive
< ETag: "67b43c4e-1a4"
< Accept-Ranges: bytes
<
apiVersion: v1
kind: ConfigMap
metadata:
        name: web-content
data:
        index.html: |
                <!DOCTYPE html>
                <html lang="en">
                <head>
                        <meta charset="UTF-8">
                        <meta name="viewport" content="width=device-width,
initial-scale=1.0">
                        <title>Simple Web App</title>
                </head>
                <body>
                        <h1>Welcome to My Simple Web Application! - Ziyan</h1>
                        <p>This application is powered by Kubernetes and MySQL.<
/p>
                </body>
                </html>
* Connection #0 to host 3.89.97.209 left intact
```

Clean up

```
ubuntu@ip-172-31-30-28:~/docker-web-mysql-app$ sudo docker-compose down
WARN[0000] /home/ubuntu/docker-web-mysql-app/docker-compose.yml: the attribute `
version` is obsolete, it will be ignored, please remove it to avoid potential co
nfusion
[+] Running 3/3
 ✔ Container docker-web-mysql-app-web-1  Removed                          0.2s
 ✔ Container docker-web-mysql-app-db-1   Removed                          1.7s
 ✔ Network docker-web-mysql-app_default  Removed                          0.1s
ubuntu@ip-172-31-30-28:~/docker-web-mysql-app$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
```

No containers are running, they are removed successfully.



Terminated my instance as well.